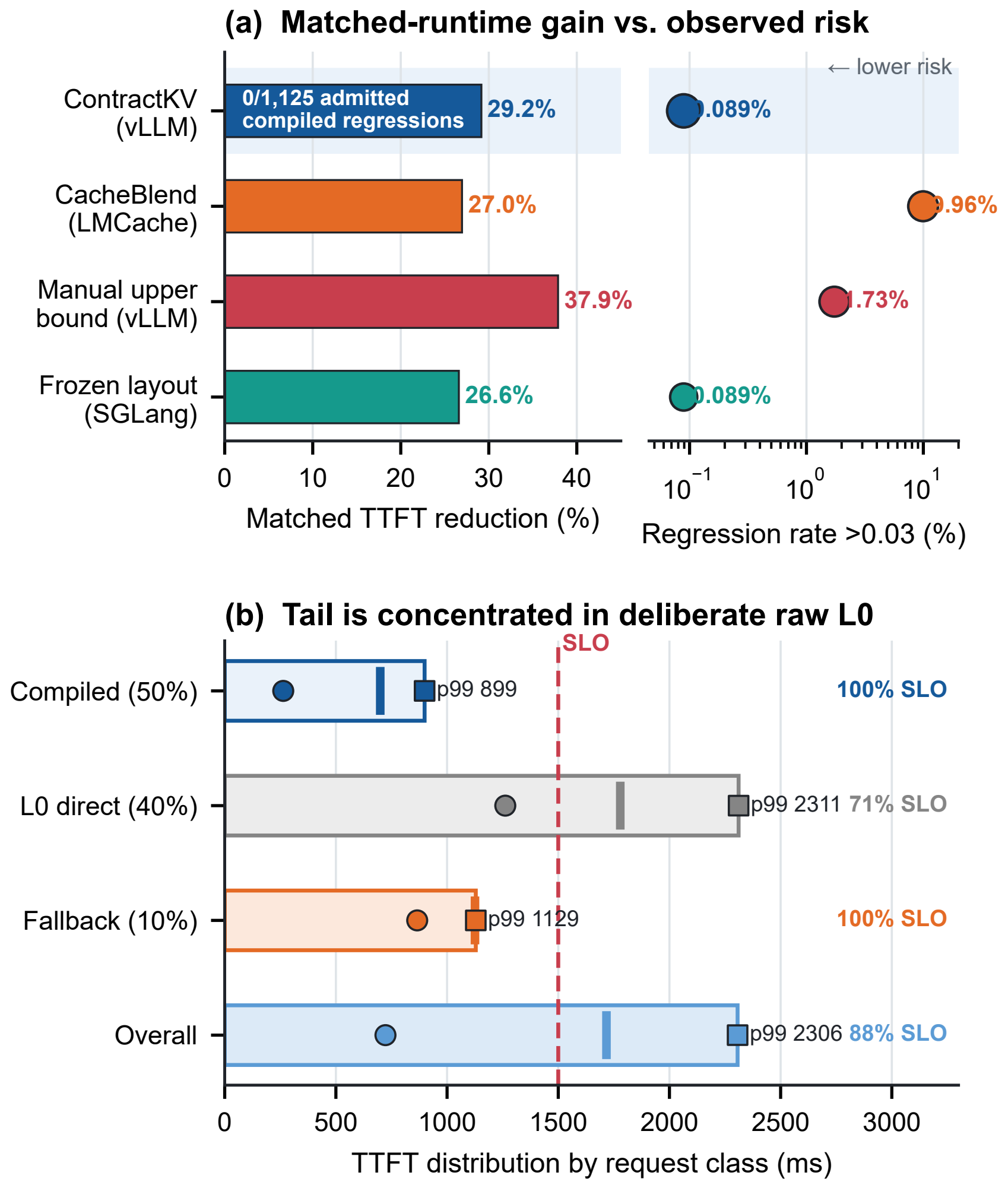

(a) Matched-runtime gain vs. observed risk
ContractKV (vLLM)
CacheBlend (LMCache)
Manual upper bound (vLLM)
Frozen layout (SGLang)
0/1,125 admitted compiled regressions
29.2%
27.0%
37.9%
26.6%
← lower risk
0.089%
0.96%
1.73%
0.089%
0
10
20
30
40
Matched TTFT reduction (%)
10^{-1}
10^{0}
10^{1}
Regression rate >0.03 (%)
(b) Tail is concentrated in deliberate raw L0
SLO
Compiled (50%)
L0 direct (40%)
Fallback (10%)
Overall
p99 899
100% SLO
p99 2311
71% SLO
p99 1129
100% SLO
p99 2306
88% SLO
0
500
1000
1500
2000
2500
3000
TTFT distribution by request class (ms)